

Spin part 4

We seek to construct spin state $|s\rangle$ from spin measurements $\langle S_x \rangle$, $\langle S_y \rangle$, and $\langle S_z \rangle$.

Let

$$x = \frac{2}{\hbar} \langle S_x \rangle$$

$$y = \frac{2}{\hbar} \langle S_y \rangle$$

$$z = \frac{2}{\hbar} \langle S_z \rangle$$

In polar coordinates

$$x = \sin \theta \cos \phi$$

$$y = \sin \theta \sin \phi$$

$$z = \cos \theta$$

By half-angle identities

$$\cos(\theta/2) = \sqrt{\frac{1 + \cos \theta}{2}} = \sqrt{\frac{1 + z}{2}}$$

$$\sin(\theta/2) = \sqrt{\frac{1 - \cos \theta}{2}} = \sqrt{\frac{1 - z}{2}}$$

Recalling that

$$\cos \phi = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\sin \phi = \frac{y}{\sqrt{x^2 + y^2}}$$

we have

$$\exp(i\phi) = \cos \phi + i \sin \phi = \frac{x + iy}{\sqrt{x^2 + y^2}}$$

Hence

$$|s\rangle = \begin{pmatrix} \cos(\theta/2) \\ \sin(\theta/2)e^{i\phi} \end{pmatrix} = \begin{pmatrix} \sqrt{\frac{1+z}{2}} \\ \sqrt{\frac{1-z}{2}} \frac{x+iy}{\sqrt{x^2+y^2}} \end{pmatrix} \quad (1)$$

Eigenmath script